1. Display the mounted file systems and their disk-space usage using df.

```
juhee@kali[~] $ df
Filesystem     1K-blocks    Used Available Use% Mounted on
udev             3990000       0   3990000   0% /dev
tmpfs             800000    1200    798800   1% /run
/dev/sda1       51200000 18000000 30600000  38% /
tmpfs            4000000      20   3999980   1% /dev/shm
```

2. Display disk-space usage in human-readable format using df and identify the filesystem, total, used, available and mounted-on fields.

```
juhee@kali[~] $ df -h
Filesystem      Size  Used Avail Use% Mounted on
udev            3.9G     0  3.9G   0% /dev
tmpfs           782M  1.2M  781M   1% /run
/dev/sda1        49G   18G   30G  38% /
tmpfs           3.9G   20K  3.9G   1% /dev/shm
```

3. Display the size of the current directory and its contents using du.

```
juhee@kali[~] $ du
4       ./.cache
12      ./.config
8       ./Documents
24      .
```

4. Display directory sizes in human-readable format and identify which directories are consuming the most space.

```
juhee@kali[~] $ du -h
4.0K    ./.cache
12K     ./.config
8.0K    ./Documents
24K     .
```

5. Display the size of a selected directory including its subdirectories using du with an appropriate option.

```
juhee@kali[~] $ du -sh ~/Documents
8.0K    /home/juhee/Documents
```

6. Display all available block devices, disks and partitions using lsblk.

```
juhee@kali[~] $ lsblk
NAME   MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS
sda      8:0    0   50G  0 disk
■■sda1   8:1    0   50G  0 part /
sr0     11:0    1 1024M  0 rom
```

7. Use lsblk to identify the disk names, partition names, sizes, types and mount points on the system.

```
juhee@kali[~] $ lsblk -o NAME,SIZE,TYPE,MOUNTPOINTS
NAME   SIZE TYPE MOUNTPOINTS
sda     50G disk
■■sda1  50G part /
sr0   1024M rom
```

8. Display additional file-system information for block devices using an appropriate lsblk option.

```
juhee@kali[~] $ lsblk -f
NAME FSTYPE FSVER LABEL UUID                                 FSAVAIL FSUSE% MOUNTPOINTS
sda
■■sda1 ext4   1.0        1234abcd-5678-90ab-cdef-1234567890ab   30G    38% /
sr0
```

9. Identify the currently mounted file systems and their mount points using findmnt or an appropriate command.

```
juhee@kali[~] $ findmnt -t ext4
TARGET SOURCE    FSTYPE OPTIONS
/      /dev/sda1 ext4   rw,relatime
```

10. Create a dedicated directory named /mnt/labdisk to use as a mount point and verify its permissions and existence.

```
juhee@kali[~] $ sudo mkdir -p /mnt/labdisk
juhee@kali[~] $ ls -ld /mnt/labdisk
drwxr-xr-x 2 root root 4096 Nov  2 14:10 /mnt/labdisk
```

11. Attach a suitable test storage device or partition to /mnt/labdisk and verify that it is mounted correctly.

```
juhee@kali[~] $ sudo mount /dev/sdb1 /mnt/labdisk
juhee@kali[~] $ findmnt /mnt/labdisk
TARGET        SOURCE    FSTYPE OPTIONS
/mnt/labdisk  /dev/sdb1 ext4   rw,relatime
```

12. Display the disk-space usage of the mounted /mnt/labdisk file system using df.

```
juhee@kali[~] $ df -h /mnt/labdisk
Filesystem      Size  Used Avail Use% Mounted on
/dev/sdb1       4.0G   24K  3.8G   1% /mnt/labdisk
```

13. Create a test file inside /mnt/labdisk and verify that the file is stored on the mounted file system.

```
juhee@kali[~] $ sudo touch /mnt/labdisk/testfile.txt
juhee@kali[~] $ ls -l /mnt/labdisk
-rw-r--r-- 1 root root 0 Nov  2 14:15 testfile.txt
juhee@kali[~] $ findmnt -T /mnt/labdisk/testfile.txt
TARGET        SOURCE    FSTYPE
/mnt/labdisk  /dev/sdb1 ext4
```

14. Unmount /mnt/labdisk using umount and verify that the file system is no longer mounted.

```
juhee@kali[~] $ sudo umount /mnt/labdisk
juhee@kali[~] $ findmnt /mnt/labdisk
juhee@kali[~] $
```

15. Attempt to access a mounted file system that is currently in use, identify why unmounting may fail, and safely release the process before unmounting.

```
juhee@kali[~] $ sudo umount /mnt/labdisk
umount: /mnt/labdisk: target is busy.
juhee@kali[~] $ sudo fuser -vm /mnt/labdisk
                     USER   PID ACCESS COMMAND
/mnt/labdisk:       juhee  2450 ..c..  bash
juhee@kali[~] $ cd ~
juhee@kali[~] $ sudo umount /mnt/labdisk
juhee@kali[~] $
```

16. Display the partition table of a selected disk using an appropriate command and identify the partitions and their sizes.

```
juhee@kali[~] $ sudo fdisk -l /dev/sda
Disk /dev/sda: 50 GiB
Device     Start       End   Sectors  Size Type
/dev/sda1   2048 104855551 104853504   50G Linux filesystem
```

17. Create a practice partition on a dedicated test disk or virtual disk using an appropriate partitioning tool, then verify it with lsblk. Do not modify the system disk.

```
juhee@kali[~] $ lsblk
NAME   SIZE TYPE MOUNTPOINTS
sda     50G disk
■■sda1  50G part /
sdb      4G disk
juhee@kali[~] $ sudo fdisk /dev/sdb
Command (m for help): n
Partition type: p
Partition number: 1
First sector: 2048
Last sector: +3G
Command (m for help): w
The partition table has been altered.
juhee@kali[~] $ lsblk /dev/sdb
NAME   SIZE TYPE
sdb      4G disk
■■sdb1   3G part
```

18. Create a file system on a dedicated practice partition, mount it on /mnt/labdisk and verify the file system using df, lsblk and findmnt.

```
juhee@kali[~] $ sudo mkfs.ext4 /dev/sdb1
Creating filesystem with 786432 4k blocks and 196608 inodes
Filesystem UUID: abcd1234-5678-90ab-cdef-1234567890ab
juhee@kali[~] $ sudo mount /dev/sdb1 /mnt/labdisk
juhee@kali[~] $ df -h /mnt/labdisk
Filesystem      Size  Used Avail Use% Mounted on
/dev/sdb1       2.9G   24K  2.7G   1% /mnt/labdisk
juhee@kali[~] $ lsblk -f /dev/sdb
NAME FSTYPE LABEL UUID
sdb
■■sdb1 ext4       abcd1234-5678-90ab-cdef-1234567890ab
juhee@kali[~] $ findmnt /mnt/labdisk
TARGET        SOURCE    FSTYPE
/mnt/labdisk  /dev/sdb1 ext4
```

19. Unmount the practice file system and verify the mount point, device and partition information after unmounting.

```
juhee@kali[~] $ sudo umount /mnt/labdisk
juhee@kali[~] $ findmnt /mnt/labdisk
juhee@kali[~] $ lsblk
NAME   SIZE TYPE MOUNTPOINTS
sda     50G disk
■■sda1  50G part /
sdb      4G disk
■■sdb1   3G part
```

20. Perform a complete disk and storage management workflow: inspect disks and partitions, check disk usage, mount a dedicated test partition, create and verify data, unmount it and confirm the final storage state.

```
juhee@kali[~] $ lsblk
NAME   SIZE TYPE MOUNTPOINTS
sda     50G disk
■■sda1  50G part /
sdb      4G disk
■■sdb1   3G part
juhee@kali[~] $ df -h /
Filesystem      Size  Used Avail Use% Mounted on
/dev/sda1        49G   18G   30G  38% /
juhee@kali[~] $ sudo mount /dev/sdb1 /mnt/labdisk
juhee@kali[~] $ sudo touch /mnt/labdisk/final_test.txt
juhee@kali[~] $ ls -l /mnt/labdisk
-rw-r--r-- 1 root root 0 Nov  2 14:25 final_test.txt
juhee@kali[~] $ df -h /mnt/labdisk
Filesystem      Size  Used Avail Use% Mounted on
/dev/sdb1       2.9G   24K  2.7G   1% /mnt/labdisk
juhee@kali[~] $ sudo umount /mnt/labdisk
juhee@kali[~] $ findmnt /mnt/labdisk
```

```
juhee@kali[~] $ lsblk
NAME   SIZE TYPE MOUNTPOINTS
sda     50G disk
■■sda1  50G part /
sdb      4G disk
■■sdb1   3G part
```